MIL-T-12664B
27 August 1965

SUPERSEDING
MIL-T-12664A(CE)
26 July 1960

MILITARY SPECIFICATION

TREATMENT, FUNGUS RESISTANT, PARANITROPHENOL, FOR CORK PRODUCTS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the treatment of cork products to render them fungus resistant.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Application. The cork products shall be treated with paranitrophenol which may be applied as a component of the bonding material formulation or in the form of a dip solution. When the

[FSC 9390]

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paranitrophenol is applied by the dip-solution method, the cork products shall be dipped after being cut to final form to insure protection of the cut edges. This method shall provide a uniform deposit without noticeable crystallization of the inhibitor on the cork.

3.2 Inhibitor content. The treatment shall deposit not less than 0.4 percent or more than 0.6 percent paranitrophenol by weight.

3.3 Fungus resistance. When tested as specified in 4.3, the treated cork product shall show no visible fungus growth.

3.4 Identification marking. After treatment, the cork products shall be permanently marked indicating treatment in accordance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Unit of product. For the purpose of inspection, the treated cork product shall be considered a unit of product.

4.2.2 Sampling.

4.2.2.1 Examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.2.2.2 Test. Sampling for test shall be in accordance with MIL-STD-105, inspection level S-4.

4.2.3 Examination. Samples selected in accordance with 4.2.2.1 shall be examined in accordance with 4.3.1. AQL shall be 2.5 percent defective.

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4.2.4 Test. Samples selected in accordance with 4.2.2.2 shall be tested as specified in 4.3.2. AQL shall be 4.0 percent defective.

4.3 Inspection procedure.

4.3.1 Examination. The treated cork shall be examined for the following defects:

101. Noticeable crystallization of the inhibitor on the cork.
102. Cork products dipped before being cut to final form.
103. Marking missing or not as specified.

4.3.2 Tests.

4.3.2.1 Paranitrophenol analysis.

4.3.2.1.1 Preparation of sample. Weigh to the nearest milligram a sample of approximately 10 grams of well shredded and thoroughly mixed particles of cork. Place the shredded sample in a thimble of a Soxhlet extractor and extract the fungicide with chloroform for not less than 3 hours. The extract shall be concentrated and then transferred quantitatively to a 100-ml volumetric flask. The solution shall then be made up to volume using chloroform, and mixed to obtain homogeneity.

4.3.2.1.2 Reagent solutions.

(a) Standard paranitrophenol solution. Weigh 1 gram of pure paranitrophenol to the nearest 0.10 milligram. Dissolve the paranitrophenol in a liter of aqueous solution of 0.05 normal sodium hydroxide and dilute 50 ml of this solution to 500 ml with distilled water. The resulting solution shall be the standard paranitrophenol solution containing 0.01 percent paranitrophenol.
(b) Titanous chloride solution. Prepare a titanous chloride solution by diluting 1 ml of a 20-percent reagent grade solution to 100 ml with boiled distilled water, cooled before use. The titanous chloride solution shall be freshly made prior to use.
(c) Borate buffer solution. Prepare a borate buffer solution by dissolving 30 grams of sodium borate decahydrate in a liter of distilled water. Adjust the pH of the resulting solution to 9.6 with a 10-percent aqueous sodium hydroxide solution.

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4.3.2.1.3 Paranitrophenol standards. Add 20 ml of a 10-percent aqueous sodium citrate solution ($Na_3C_6H_5O_72H_2O$) to each of nine 100-ml volumetric flasks. Use eight of the flasks to prepare standards equivalent to samples containing the following amounts of paranitrophenol expressed in milligrams: 0.20, 0.25, 0.30, 0.40, 0.50, 0.60, 0.70, and 0.80 by adding 2, 2.5, 3, 4, 5, 6, 7, and 8 ml of standard paranitrophenol solution, respectively.

4.3.2.1.4 Preparation of unknown. Add 1 ml of the sample prepared in 4.3.2.1.1 to the remaining flask containing the sodium citrate solution described in 4.3.2.1.3.

4.3.2.1.5 Procedure. The standards prepared as specified in 4.3.2.1.3 and the unknown prepared as specified in 4.3.2.1.4 shall be treated alike in the subsequent procedure. Reduce the paranitrophenol to paraaminophenol by slowly adding titanous chloride solution until the color changes from yellow to slight purple. This reduction requires from 5 to 10 ml titanous chloride solution. Add 30 ml of a borate buffer solution. Add 10 ml of a 5-percent aqueous phenol solution and shake vigorously. Make to volume with the buffer solution and mix thoroughly. Let the solution stand for 1 to 3 hours in subdued light to develop color. Determine the concentration in milligrams of paranitrophenol in the unknown by one of the following methods:

(a) Spectrophotometer - Read the colors of the standards at 630 millimicrons and plot the curve. Read the unknown and obtain the milligrams of paranitrophenol from the standard curve.
(b) Electrophotometer - Read the color of the standards and unknown using a filter showing maximum transmission at about 630 millimicrons. Plot the standard curve and obtain the concentration as specified for the spectrophotometer.
(c) Visual - Transfer standards and unknown to 18- by 150-mm test tubes, and match the color of the unknown to the standard. View the tubes against a white background.

4.3.2.1.6 Calculations. Obtain the percent paranitrophenol as follows:

$$\text{percent paranitrophenol} = \frac{A \times 10^2 \times 100}{B}$$

Where:

A = weight in grams of paranitrophenol contained in aliquot.
B = weight in grams of the sample.

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4.3.2.1.7 Failure. Evidence of a deposit of less than 0.4 percent or more than 0.6 percent paranitrophenol by weight shall constitute failure of this test.

4.3.2.2 Fungus resistance. Determination of the fungus resistance shall be made by the following microbiological test method.

4.3.2.2.1 Test organisms. Treated cork products shall be tested using the following fungi (see 6.3):

	American Type Culture Collection Number	U. S. Army Natick Laboratories Number
Aspergillus niger	9642	QM386
Aspergillus flavus	9643	QM380
Penicillium funiculosum	9644	QM391
Trichoderma sp.	9645	QM365

4.3.2.2.1.1 Maintaining stock cultures. Cultures of these fungi shall be maintained separately on a potato-dextrose agar medium. The stock cultures may be kept for not more than 4 months in a refrigerator at approximately 3° to 10° C. Subcultures incubated at 28° to 30° C. for 7 to 20 days shall be used in preparing the inoculum.

4.3.2.2.2 Inoculum. The inoculum shall be prepared by pouring into one subculture of each of the four fungi in a ripe fruiting condition approximately 10 ml of sterile water containing about 0.005 percent of a nontoxic wetting agent. Force the fungus spores into suspension. (The use of a sterile camel's hair brush is suggested as a means of preparing this suspension.) The four spore suspensions shall be combined under sterile conditions, thoroughly mixed, and then diluted to 200 ml with sterile distilled water. The composite spore suspension shall be prepared fresh each day or shall be held in a refrigerator at a temperature of 3° to 10° C. for not more than 7 days.

4.3.2.2.3 Procedure and evaluation of results.

4.3.2.2.3.1 Test agar medium. The medium shall have the following composition:

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Distilled water	1,000 ml
NH_4NO_3	3.0 gm
KH_2PO_4	1.0 gm
$MgSO_4.7H_2O$	0.5 gm
KCl	0.25 gm
Agar	15.0 gm

The pH shall be 5.5 to 6.5; if otherwise, it shall be adjusted to that range with HCl or NaOH. After mixing, the above ingredients shall be sterilized by autoclaving for 15 minutes at 15 psi (121° C.). Under sterile conditions, the medium shall be poured into 10-cm Petri dishes, about 30 ml per dish, and allowed to harden.

4.3.2.2.3.2 Test specimens. Four treated cork specimens shall be completely immersed in slowly running water for 24 hours and subsequently air dried. The leaching tank shall be large enough to hold an amount of water weighing not less than 50 times the weight of the specimens. The water shall not fall directly on the specimens and shall flow at a rate of 5 to 10 liters per hour. The pH of the water shall be in the range of 6.0 to 8.0.

4.3.2.2.3.3 Inoculation. Under aseptic conditions, each test specimen shall be immersed in distilled water and placed firmly on the surface of the solidified agar medium contained in the 10-cm Petri dishes. Distribute 1.0 to 1.5 ml of the composite inoculum over the surface of the specimen and surrounding medium. The use of a sterile pipette is suggested as one means of inoculating the test specimens.

4.3.2.2.3.4 Incubation. The period of incubation shall be 14 days at a temperature of 28° to 30° C. and a relative humidity of 85 to 95 percent.

4.3.2.2.3.5 Controls. Three untreated specimens of cork shall be tested with the treated specimens to check the viability of the inoculum. At the end of the incubation period, the controls shall be well covered with the fungus growth.

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4.3.2.2.4 Failure. Upon completion of the incubation period, the test specimens shall be examined. The treatment shall be considered to have failed to meet the requirements of this specification when any of the test specimens show visible evidence of fungus growth except that growth on the edges of test specimens that must be cut to suitable size for testing shall be ignored.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Intended use. The paranitrophenol treatment is intended to render cork products, such as cork gaskets and cork sheet, fungus resistant. Cork products which come in contact with a person's skin during prolonged use or with food and liquids to be consumed by humans should not be treated by this method.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

6.3 Test organisms. The organisms used in the fungus-resistance test specified in 4.3.2.2.1 may be obtained from the American Type Culture Collection, 12301 Parklawn Drive, Rockville Maryland, 20852 or, for Service use, from the Pioneering Research Division, U. S. Army Natick Laboratories, Natick, Mass.

6.4 A suggested composition of the fungusproofing treating solution is as follows:

Paranitrophenol	.2 percent minimum
Isopropyl alcohol	10 percent minimum
Stoddard solvent	88 percent maximum

Custodians:

Army - MO(ERDL)
Air Force - 69

Preparing Activity:

Army - MO(ERDL)

Review activities:

Army - GL, MU, MD, MO
Air Force - 69

User activity:

Army - MI, EL
Air Force - None

Project No. 9390-0012

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-12664B; Treatment, Fungus Resistnat, Paranitrophenol, for Cork Products

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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